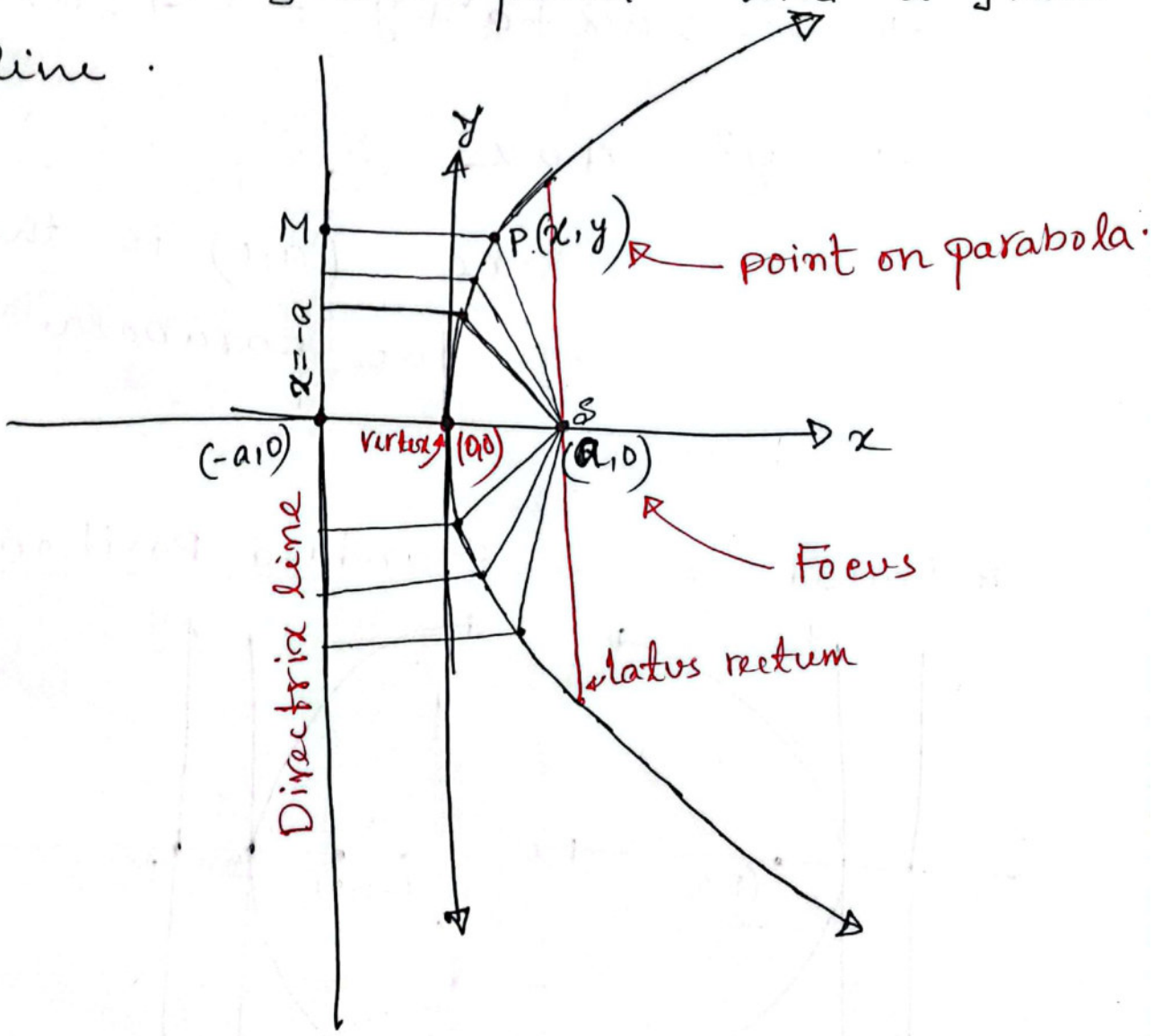
Lec - 12

conic sections (James Stewart - calculus - 10.5)

* Parabola:

A parabola is the set of all points in the plane that are equidistant from a fixed point and a fixed line.



According to the definition.

$$SP = ePM \quad ; e = 1$$

$$\Rightarrow \sqrt{(x-a)^2+(y-0)^2} = \left| \frac{x+a}{\sqrt{1^2+0^2}} \right|$$

$$\Rightarrow \sqrt{(x-a)^2+y^2} = \pm \frac{(x+a)}{1} = \pm (x+a)$$

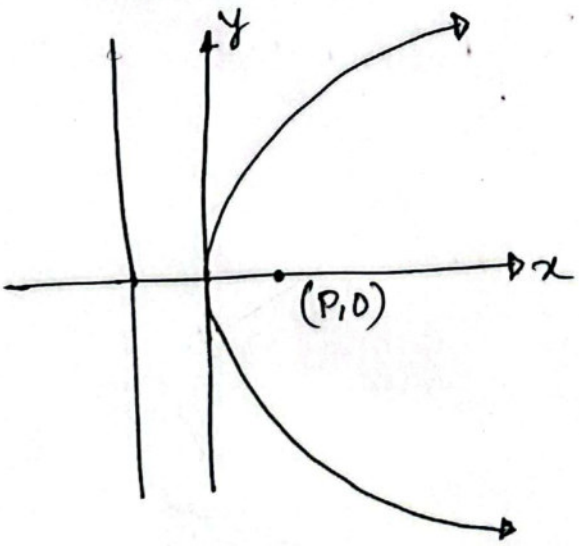
$$\Rightarrow (x-a)^2+y^2 = (x+a)^2$$

$$\Rightarrow x^2-2ax+a^2+y^2 = x^2+2ax+a^2$$

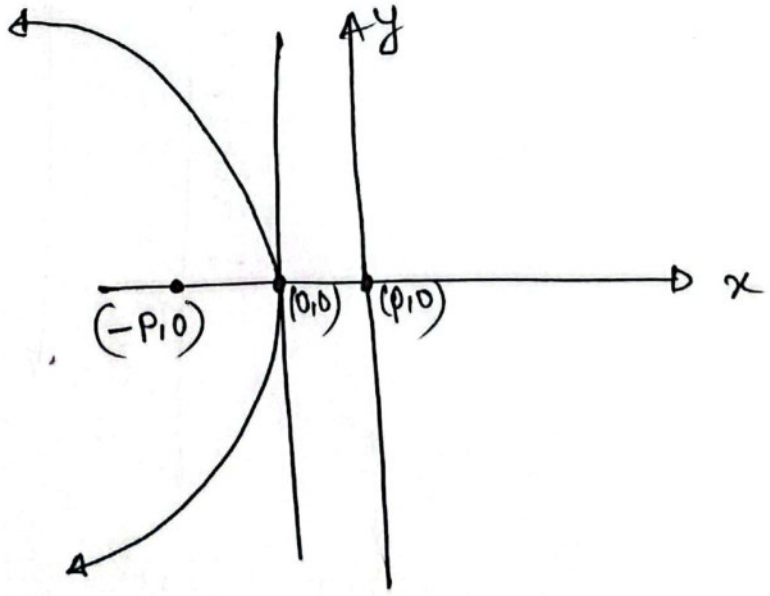
$$\Rightarrow y^2 = 4ax$$

where $(a,0)$ is the Focus of the parabola $y^2 = 4ax$

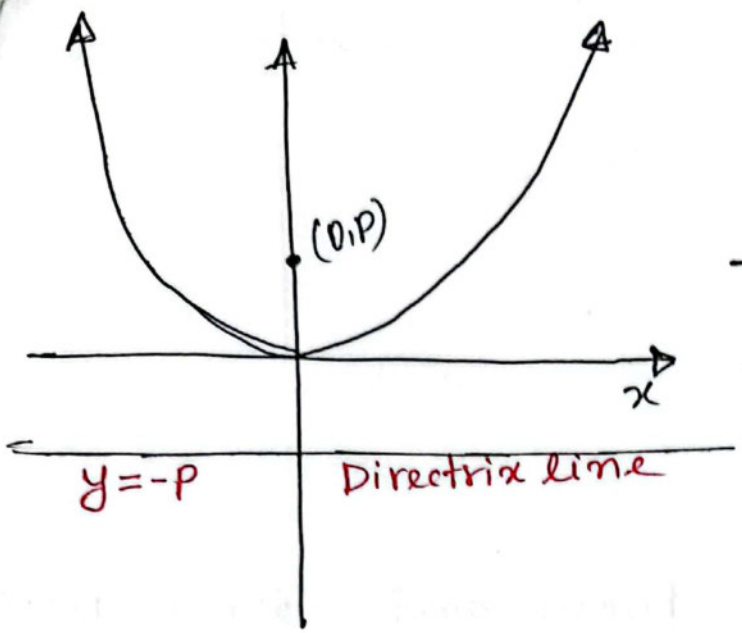
☆ Parabolas in Standard Position:



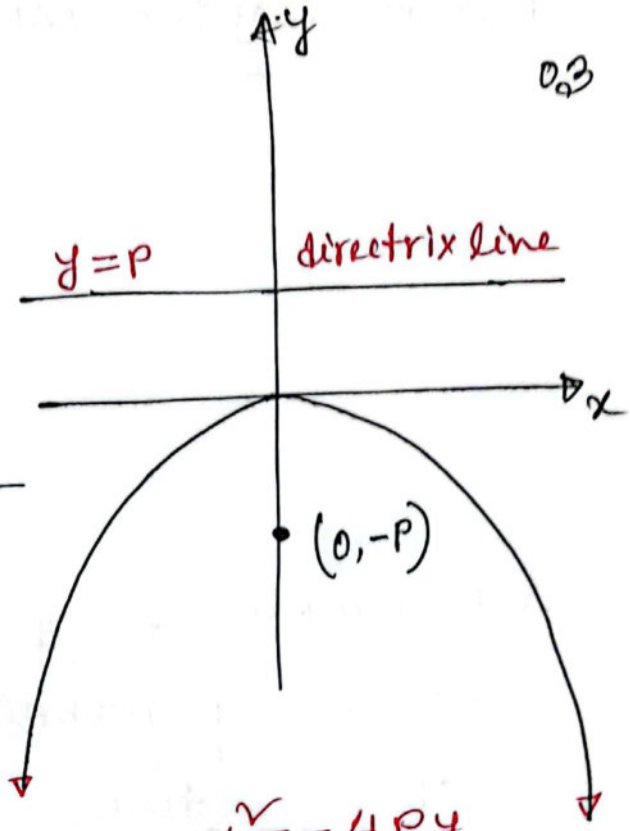
$y^2 = 4px$



$y^2 = -4px = 4(-p)x$

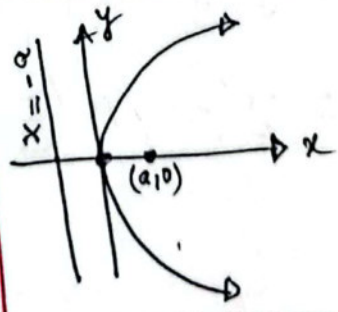
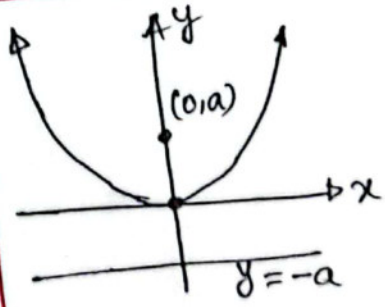


$x^2 = 4py$



$x^2 = -4py$

★ Necessary formulae related to the parabola $y^2 = 4ax$ and $x^2 = 4ay$.

Description.	$y^2 = 4ax$	$x^2 = 4ay$.
		
1) co-ordinates of the Vertex	$x=0, y=0$ $(0,0)$	$x=0, y=0$ $(0,0)$
2) co-ordinates of Focus	$x=a, y=0$ $(a,0)$	$x=0; y=a$ $(0,a)$
3) Equation of directrix	$x=-a$	$y=-a$
4) Equation of the latus rectum.	$x=a$	$y=a$.

5) Equation of the axis	$y=0$	$x=0$

★ <u>Example</u>: Find Focus and directrix of the parabola ~~y~~ $y^2+10x=0$ and sketch the graph.

<u>Ans:</u>

$$y^2+10x=0$$

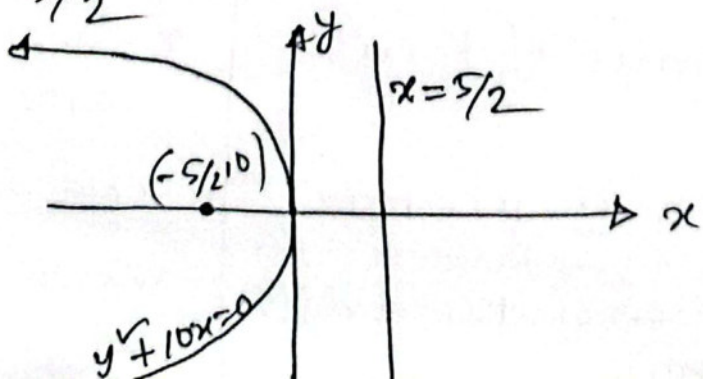
$$\Rightarrow y^2=-10x$$

$$\Rightarrow y^2=-4\cdot\frac{10}{4}x$$

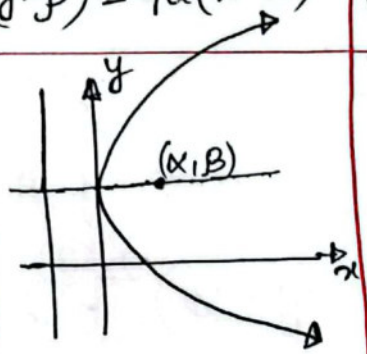
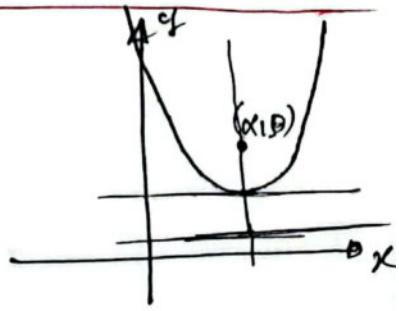
$$\Rightarrow y^2=4\left(\frac{-5}{2}\right)x.$$

$y^2=4px$ where $p<0$

Focus at $\left(-\frac{5}{2},0\right)$ and directrix is $x=5/2$



★ Necessory formulae related to the parabola $(y-\beta)^2 = 4a(x-\alpha)$ & $(x-\alpha)^2 = 4a(y-\beta)$

	$(y-\beta)^2 = 4a(x-\alpha)$	$(x-\alpha)^2 = 4a(y-\beta)$
consider $(y-\beta)^2 = 4a(x-\alpha)$ as $Y^2 = 4aX$, where $Y = y-\beta$ and $X = x-\alpha$. then use previews standard Formulae.		
1) co-ordinates of the vertex	$X=0$; $Y=0$ $\Rightarrow x-\alpha=0$; $y-\beta=0$ $\Rightarrow x=\alpha$; $y=\beta$ (α,β)	$X=0$; $Y=0$ $\Rightarrow x-\alpha=0$; $y-\beta=0$ $\Rightarrow x=\alpha$; $y=\beta$ (α,β)
2) Co-ordinates of Focus	$X=a$; $Y=0$ $\Rightarrow x-\alpha=a$; $y-\beta=0$ $\Rightarrow x=a+\alpha$; $y=\beta$ $(a+\alpha,\beta)$	$X=0$; $Y=a$ $\Rightarrow x-\alpha=0$; $y-\beta=a$ $\Rightarrow x=\alpha$; $y=a+\beta$ $(\alpha, a+\beta)$
3) Equation of directrix	$X=-a$ $\Rightarrow x-\alpha=-a$ $\Rightarrow x=-a+\alpha$	$Y=-a$ $\Rightarrow y-\beta=-a$ $\Rightarrow y=\beta-a$
4) Equation of the latus rectum.	$X=a$ $\Rightarrow x-\alpha=a$ $\Rightarrow x=\alpha+a$	$Y=a$ $y-\beta=a$ $y=\beta+a$
5) Equation of the axis	$Y=0$ $\Rightarrow y-\beta=0$	$X=0$ $\Rightarrow x-\alpha=0$

*Example: Describe the graph of the equation $y^2 - 8x - 6y - 23 = 0$.

Ans:

$$y^2 - 8x - 6y - 23 = 0$$

$$\Rightarrow y^2 - 2\cdot 3\cdot y + 3^2 - 3^2 - 8x - 23 = 0$$

$$\Rightarrow (y-3)^2 - 8x - 32 = 0$$

$$\Rightarrow (y-3)^2 = 8x - 32$$

$$\Rightarrow (y-3)^2 = 8(x-4)$$

$$\Rightarrow (y-3)^2 = 4\cdot 2(x-4) \longrightarrow (1)$$

$$\Rightarrow \cancel{Y^2 = 4\cdot aX}$$

compare (1) with $Y^2 = 4aX$

Here, $Y = y-3$; $X = x-4$; $a = 2$

- Vertex: $X = 0$; $Y = 0$

$\Rightarrow x-4 = 0$; $y-3 = 0$

$\therefore x = 4$; $y = 3$

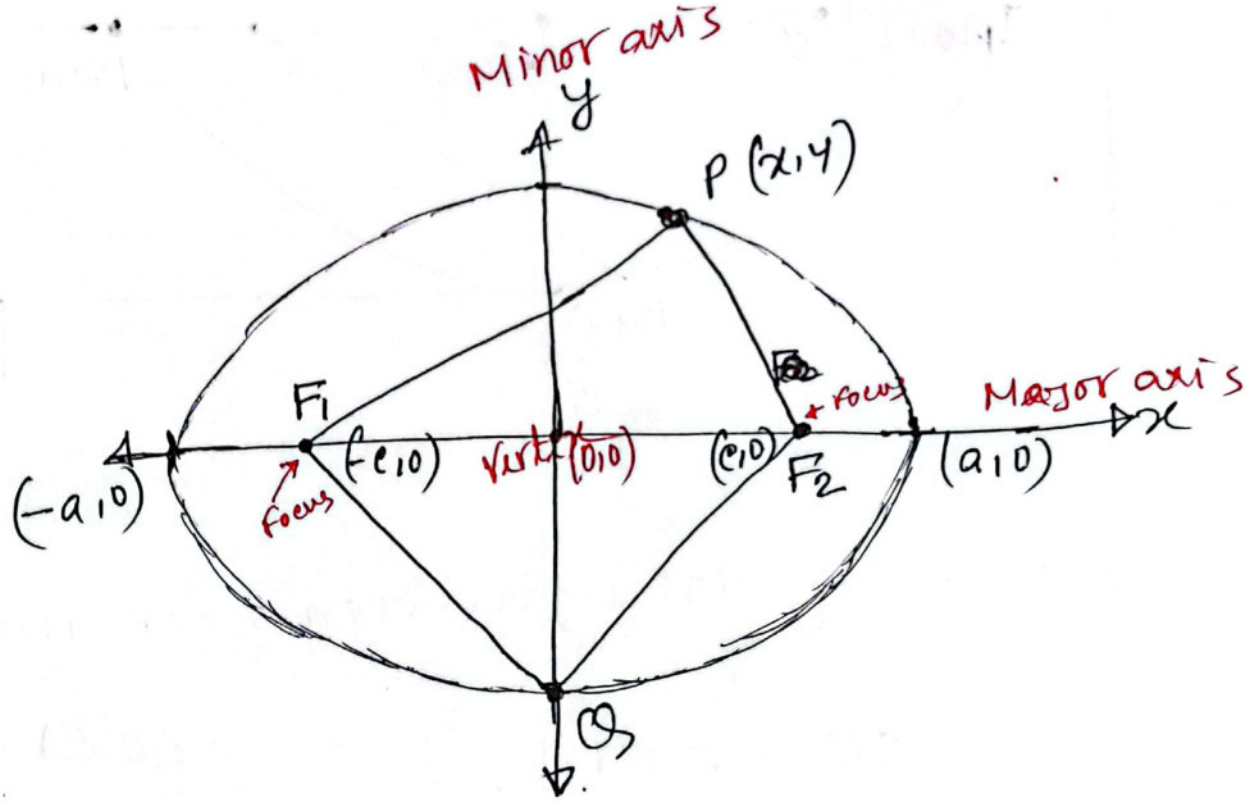
$(4, 3)$

- Focus: $X = a = 2$; $Y = 0$

$\Rightarrow x-4 = 2 \Rightarrow x = 6$; $y-3 = 0$; $y = 3$ $(6, 3)$

* Ellipse:

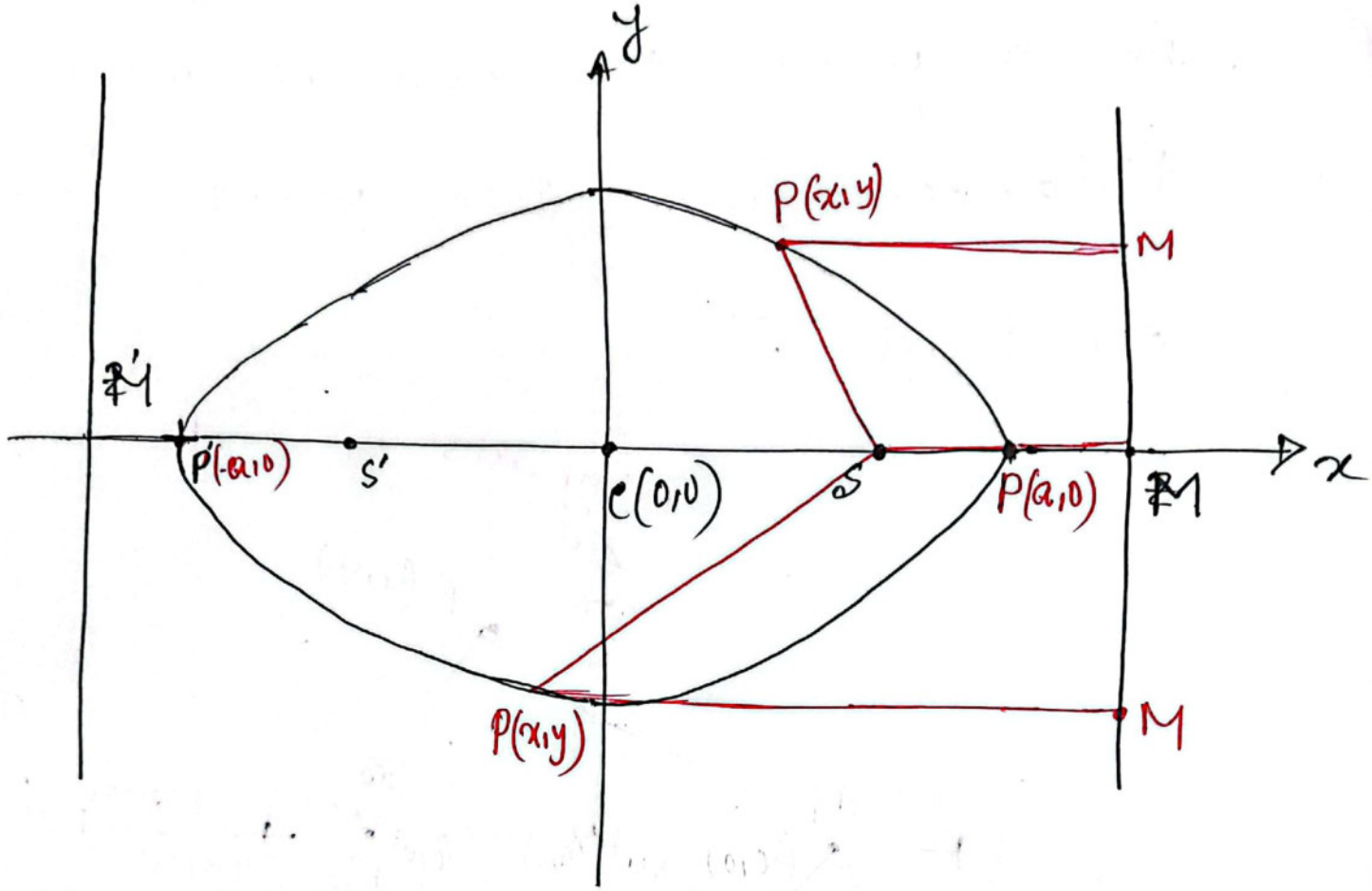
An ellipse is a the set of points in a plane the sum of whose distances from two points is a constant.



$$F_1P + F_2P = F_1Q + F_2Q = 2a.$$

(08)

★ standard form derivation of Ellipse:



According to definition for right directix

$$SP = ePM \quad ; \; 0<e<1$$

$$\Rightarrow CP - CS = e(CM - CP)$$

$$\Rightarrow a - CS = e(CM - a) \longrightarrow (1)$$

For ~~Left directrix~~, S, P' and ~~Z~~

$$SP' = e\,P'M$$

$$\Rightarrow a + es = e(a + eM) \rightarrow (2)$$

Now, (1) + (2)

$$a - es = e(eM - a)$$
$$a + es = e(a + eM)$$
$$\overline{2a = e(2eM)}$$

$$\Rightarrow eM = \frac{a}{e}.$$

and, (2) - (1)

$$a + es = e(a + eM)$$
$$a - es = e(eM - a)$$
$$(-) \quad (+) \qquad (-)$$
$$\overline{2es = e(a + eM - eM + a)}$$

$$\Rightarrow 2es = 2ae$$

$$\therefore es = ae,$$

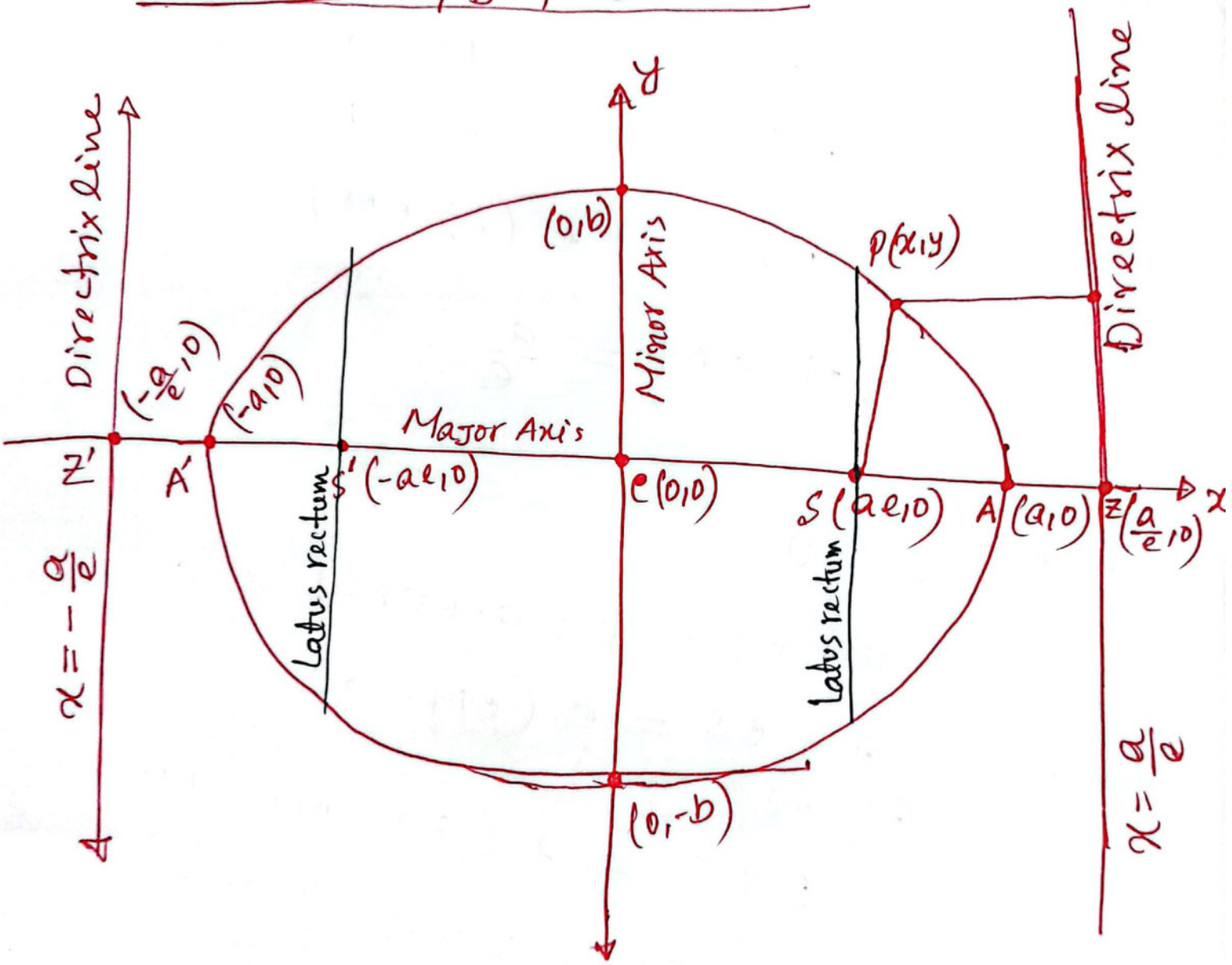
So, Foeuses / Foci are $(\pm ae, 0)$

Equations of directrix: $x = \pm \frac{a}{e}$.

CamScanner

★ Standard form of Ellipse:

Form-01 : $a > b$; $0 < e < 1$



· Equation of ellipse when $a > b$.

$$SP = e\,PM \quad ; \quad 0 < e < 1$$

$$\Rightarrow SP^2 = e^2 PM^2$$

$$\Rightarrow (x - ae)^2 + (y - 0)^2 = e^2 \left| x - \frac{a}{e} \right|^2$$

(1)

$$\Rightarrow x^2 - 2aex + a^2e^2 + y^2 = e^2\left(x^2 - 2\frac{a}{e}x + \frac{a^2}{e^2}\right)$$

$$\Rightarrow x^2(1-e^2) + y^2 = a^2 - a^2e^2$$

$$\Rightarrow x^2(1-e^2) + y^2 = a^2(1-e^2)$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad ; \text{let, } b^2 = a^2(1-e^2)$$

$$\Rightarrow \frac{b^2}{a^2} = 1 - e^2$$

$$\Rightarrow e^2 = 1 - \frac{b^2}{a^2}$$

$$\Rightarrow e = \sqrt{\frac{a^2-b^2}{a^2}}$$

S, S' = Focuses.

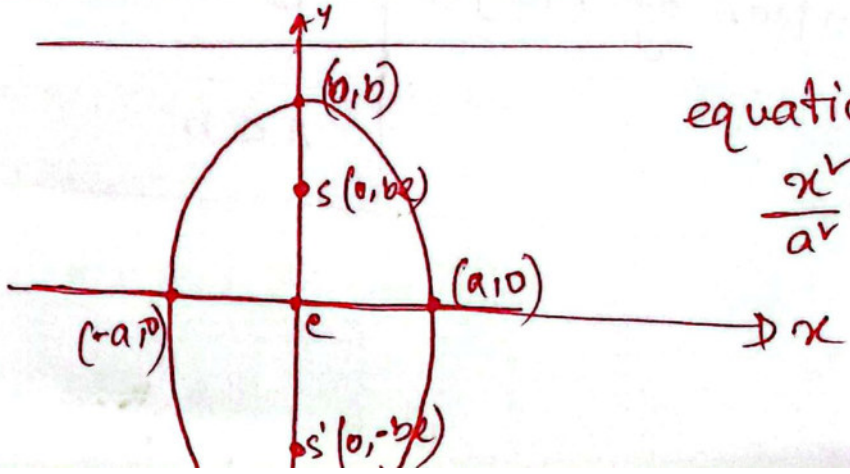
C = centore.

A, A' = Vertices.

SS' = Major Axis

e = Eccentricity

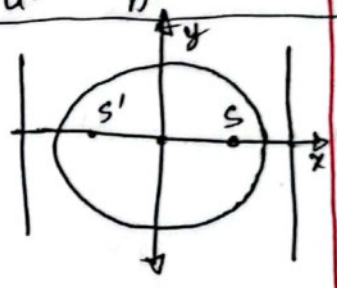
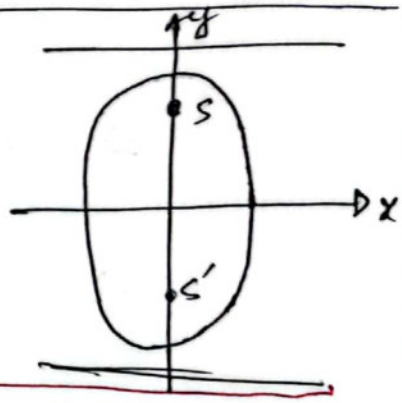
* Form-02 : $a < b$; $0 < e < 1$:



equation

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad ; b > a$$

★ Necessary Formulae related to ellipse ★

	$a > b$	$b > a$
	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
		
(1) centre	$x=0, y=0$	$x=0, y=0$
(2) Vertices	~~$x=\pm ae, y=0$~~ $x=\pm a; y=0$	~~$x=0, y=\pm be$~~ $x=0; y=\pm b$
(3) co ordinates of Foci / Focuses	$x=\pm ae, y=0$	$x=0; y=\pm be$
(4) Equations of Directrix	$x=\pm a/e$	$y=\pm b/e$
(5) Equations of Latus rectum	$x=\pm ae$	$y=\pm be$
(6) Equation of Major Axis	$y=0$	$x=0$
(7) Area	πab	πab

CamScanner